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RECORD OF CHANGES

Rev.	Date	Author	Description
0.0	05/08/2013	JLDECARLOS	Initial version

1 AIM

The purpose of the present document is to explain the content included in the 2MW and 2.5MW wind turbine cable layouts.

2 SCOPE

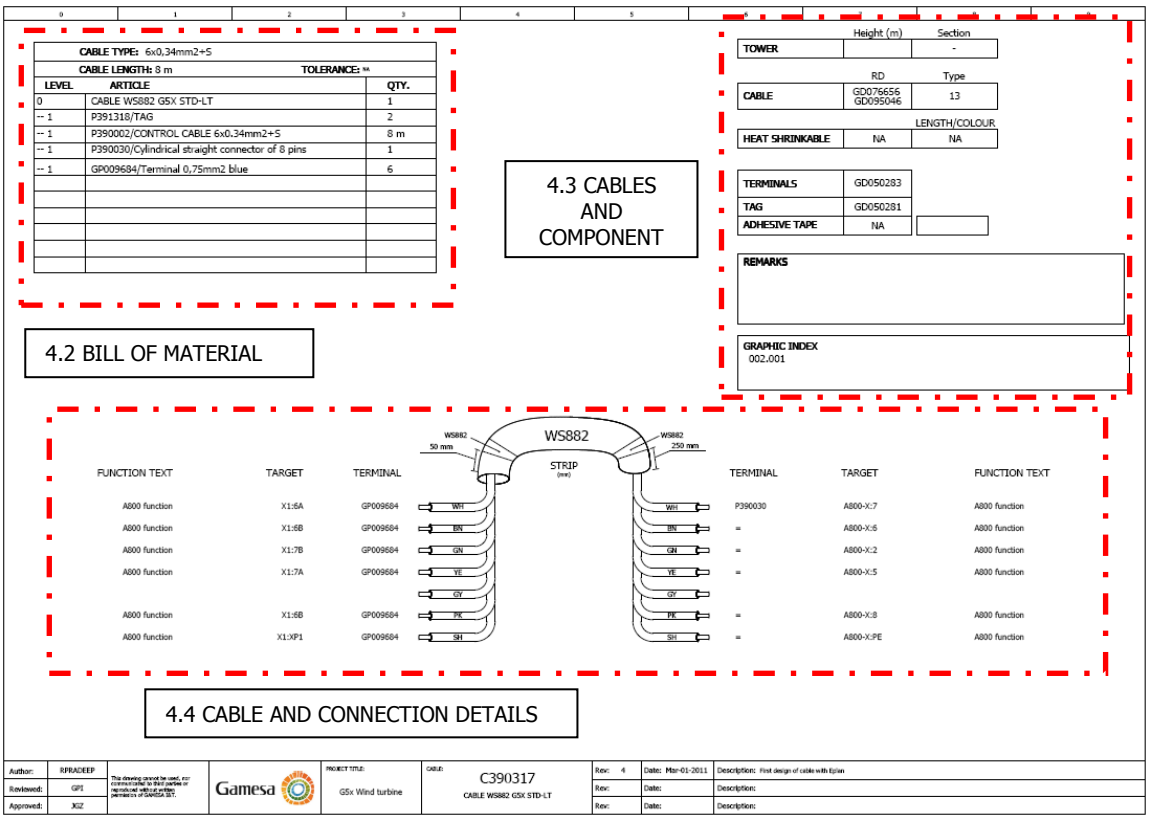
This document applies to all 2MW and 2.5MW wind turbine grouping cables, power cables, fibre optic cables, earthing cables, power supply cables and signal cables.

3 DEFINITIONS AND ACRONYMS

Not Applicable.

4 DESCRIPTION

Each GAMESA cable grouping will include in its documental structure a cable layout showing the bill of materials of the grouping, the codes of the GAMESA requirements documents of the cable and components, any other document to be applied, the parts codes, the configuration of the connector and any useful or relevant information to manufacture and/or to install the cable. The following pictures intend to show an example of such kind of cable layout.



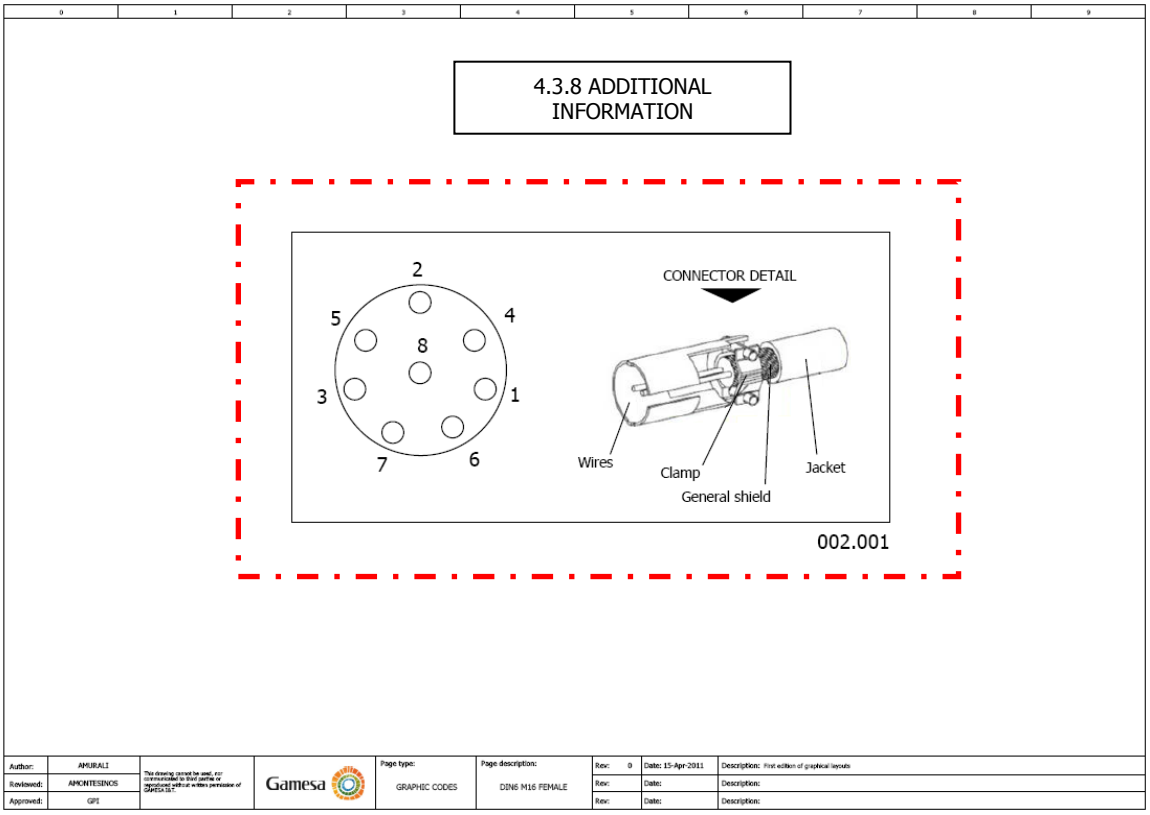


Figure 1. Cable layout sample

4.1 DEFINITIONS AND ACRONYMS USED IN THE CABLE LAYOUT

NA	Not Applicable
RD	Requirements Document
REV	Cable Layout revision

Table 1. Definitions and Acronyms used in the cable layout

4.2 BILL OF MATERIAL

The list of materials and the codes of the components that the cable grouping contains is shown in this section of the cable layout. The cable grouping is normally composed by a cable, tags, terminals, connectors…

4.2.1 CABLE TYPE

It indicates the cable type, section, number of cores…

4.2.2 CABLE LENGTH AND TOLERANCE

Cable length and tolerance required to the supplier are indicated. If no tolerance is specified "NA" (Not Applicable) will be written.

4.3 CABLE AND COMPONENTS INFORMATION

4.3.1 TOWER

Tower height and section of the tower where the cable belongs to is shown in this field.

NUMBER OF SECTIONS OF THE TOWER	SECTION DESIGNATION
2	BOTTOM
	TOP
3	BOTTOM
	MIDDLE
	TOP
4	BOTTOM
	LOWER MIDDLE
	UPPER MIDDLE
	TOP

Table 2. Designation of the sections of different towers

If the cable does not belong to the subsystem "tower" or it is not specific of one particular tower height a dash is placed in these fields.

4.3.2 CABLE

The requirements documents codes of the cable are indicated in the column named "RD". The number that appears in the "Type" column makes reference to the cable features combination taken from the requirements document of the cable.

4.3.3 HEAT SHRINKABLE

The requirements document code of the heat shrinkable used in the cable is shown, if no heat shrinkable is used "NA" (Not Applicable) will be placed.

4.3.4 TERMINALS

The requirements documents codes of the terminals and connectors used in the cable are indicated.

4.3.5 TAG

The requirements document code of the tag used in the cable is shown.

4.3.6 ADHESIVE TAPE

If an adhesive or insulating tape is used, the specification for such material will be included. The distance in mm to place the tape is also shown.

4.3.7 REMARKS

Any special instruction to be carried out, special item or explanation to the cable layout, any new document to be applied, comment or interesting information will be indicated in this field.

4.3.8 ADDITIONAL INFORMATION CODES

If extra or additional information to the one included in "REMARKS" is needed, the cable layout will have one or several additional sheets to show such information.

It can indicate the configuration of a connector, the position of the pins, a special connection procedure, a significant picture or any relevant information that might be required.

This information will be included in a new sheet with the same code as the one shown in the cable layout "Additional information codes" section.

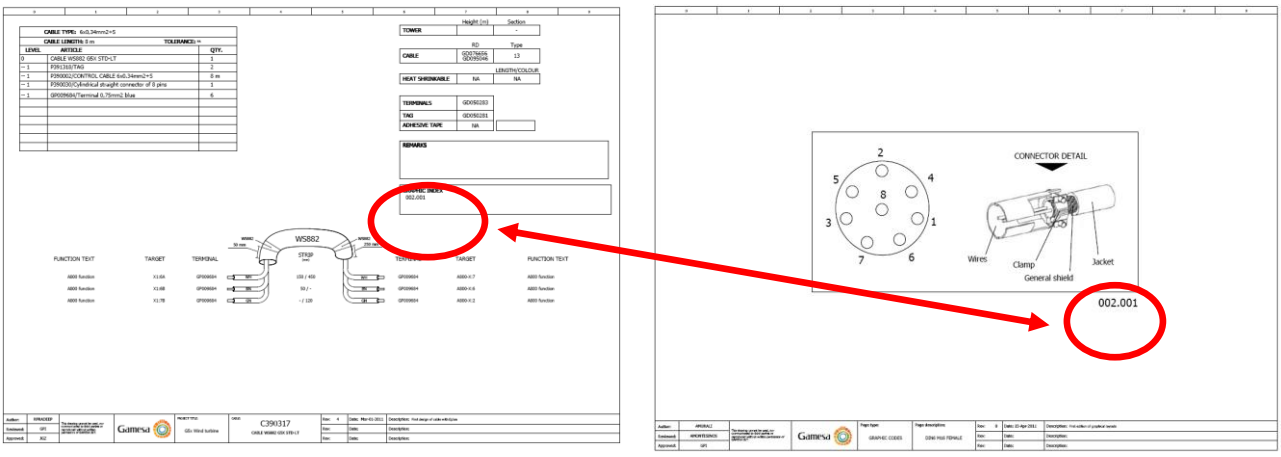


Figure 2. Additional information code sample

In the sample above a picture of the connector used in the cable is needed to clarify the pins position, so an additional sheet is created. The code in the section "Additional information codes" is used to identify, in the new sheet, the information required.

4.4 CABLE AND CONNECTION DETAILS

4.4.1 FUNCTION TEXT

It represents the function of the cable or the system where it belongs to.

4.4.2 TARGET

It represents the target of each core of the cable, where it is connected to and in case of a cable with a connector with more than one pin, the number of the pin to connect each one of the cores.

If the cable contains a connector, the letter "X" will be used to designate the connector, if there is more than one connector in the same side of the cable, a number will be added, (X1, X2…). The pin number of the connector, where each one of the cable cores will be connected, is also indicated after the colon.

DESIGNATION	EXPLANATION
A800-X: 7	A800 is the target device
	X means there is one connector
	7 indicates the PIN 7 of the X connector
X1: 6A	X1:6A is the target device, no connector is used
A800-X2: 6	A800 is the target device
	X2 means there is more than one connector (at least two)
	6 indicates the PIN 6 of the X2 connector

Table 3. Target code explanation

4.4.3 TERMINAL

This field indicates the terminal, connector or cable lug used in each core of the cable, in case there is one connector for more than one core of the same cable the equal sign is placed instead the code.

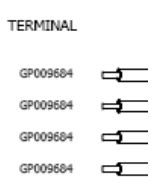
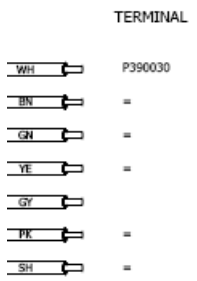
SAMPLES	
	
One terminal per core	One connector connected to six cores

Figure 3. Terminals and connectors samples

4.4.4 CABLE SCHEME

The cable is represented by means of a schematic figure with the tag name, the position of the tags, the cores and its colours. The shield is represented as another core.

If the stripping distance of the outer sheath of the cable needs to be specified it will be indicated below the column STRIP. This stripping distance can be shown for one or for both ends of the core.

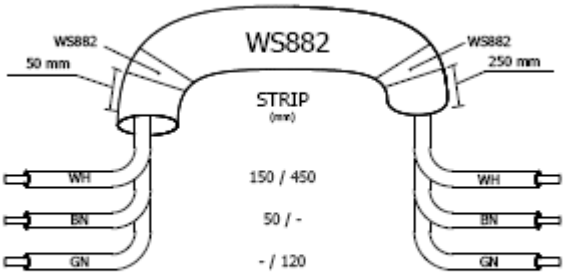


Figure 4. Cable scheme sample

5 REFERENCED DOCUMENTS

This document is equivalent to the GD128248

GD128248 RD Cable layout interpretation G5x